

Quiz - Monod's Nightmare

My name is: _____

Under ideal growing conditions in the laboratory, the population size of the common bacterium *Escherichia coli* can double in around 20 minutes. *E. coli*'s cells are rod-shaped and are roughly approximated by a rectangular box that is $2\ \mu\text{m}$ long, $1\ \mu\text{m}$ wide, and $1\ \mu\text{m}$ high. ($1\ \mu\text{m} = 1\ \text{micro-meter} = 1 \times 10^{-6}\ \text{meters} = 0.000001\ \text{m}$). For comparison, a strand of human hair is roughly $100\ \mu\text{m}$ wide.

(a) Would a continuous-time model (the solution for which is $N_t = N_0 e^{rt}$) or a discrete-time model (the solution for which is $N_t = \lambda^t N_0$) be more appropriate for describing this population? Why?

(b) What is *E. coli*'s growth rate r (in minutes) under these ideal growing conditions. Express your answer by solving for the equation you would use to calculate r to the simplest solution possible.

(c) Let's assume that our classroom is roughly $10\ \text{m}$ long, $10\ \text{m}$ wide, and $8\ \text{m}$ high (it's not, I just made up these numbers). How long it would take for an exponentially growing population of *E. coli* under ideal conditions to fill our empty classroom when starting from a single individual bacterium?